SmartCam: A Collaborative Multi-Agent SLAM Framework for Autonomous Navigation

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***Abstract*–Autonomous vehicle navigation in large-scale urban and indoor environments faces critical challenges from GPS unavailability, cumulative localization errors, and sensor failures that could potentially compromise safety. Current solutions rely heavily on GPS modules or centralized mapping systems that struggle with scalability and robustness. We propose SmartCam, a collaborative SLAM framework where multiple cars share local submaps at common landmarks to build globally consistent maps. The target architecture uses a central server to coordinate vehicle status and future task allocation. The system itself integrates a sensor-based emergency braking layer for fault tolerance. This report presents ROS 2 communication and SLAM component development alongside simulation-based learning work. The VLA contribution comprises Gazebo demonstration capture, LeRobot conversion, and a 100-step BF16 SmolVLA run; evaluation on nine training-episode frames measures offline command imitation rather than autonomous navigation or physical deployment.**

***Index Terms*—Autonomous Driving, Computer Vision, Path Planning, Embedded AI, Smart Mobility, Edge Computing, Reinforcement Learning.**

I. Introduction

The paradigm shift in autonomous mobile robotics from monolithic single-agent systems to decentralized Multi-Agent Systems (MAS) has unlocked unprecedented capabilities in collective environmental perception and operational redundancy. However, realizing robust Simultaneous Localization and Mapping (SLAM) within such frameworks poses significant computational challenges, particularly when deployed on resource-constrained edge devices [1]. The primary bottleneck lies in the high-fidelity processing of asynchronous sensor data, spanning LiDAR point clouds, IMU inertial frames, and visual streams, that must be synthesized in real time to ensure trajectory coherence.

Three fundamental challenges motivate our work. First, GPS is unavailable in many indoor and dense urban environments. Second, odometric position drift accumulates over time, causing maps to degrade without correction. Third, single-sensor architectures are fragile in the face of sensor failure. These constraints motivate sensor fusion and collision-aware local planning [2], [3].

SmartCam combines local SLAM, sensor fusion, and a centralized map-merging design on low-cost mobile robots. Figure 1 summarizes the target architecture. Reinforcement learning (RL) investigates navigation policies, while the separate Vision-Language-Action (VLA) branch investigates image- and language-conditioned velocity prediction from demonstrations. The VLA work reported here is a workstation-based simulation and offline feasibility study; it has not been integrated into autonomous robot navigation.

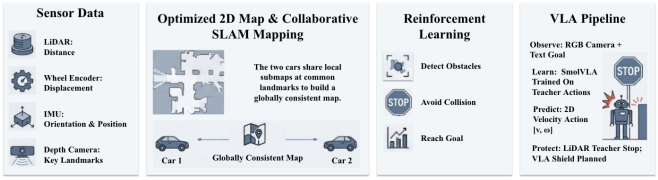


Fig. 1: Target SmartCam architecture. SLAM, map merging, RL, and VLA are development branches; the diagram does not establish an integrated autonomous system.

II. Methodology

A. SLAM Toolbox and Multi-Agent Map Merging

For the multi-agent deployment of the SmartCam fleet across the Rice University campus, a centralized 2D map-merging architecture was selected to balance edge-compute constraints with robust navigation capabilities. To overcome the strict NAT and multicast restrictions in the university's wireless infrastructure, the communication pipeline uses a Tailscale virtual private network and the Zenoh protocol. This architecture ensures secure, low-latency, unicast data transfer between the peripheral rover units (equipped with Raspberry Pi 4s) and the central mapping server (hosted on a Jetson Orin Nano).

Rather than saturating network bandwidth with dense 3D point clouds, each edge unit performs local simultaneous localization and mapping (SLAM) to generate highly compressed 2D occupancy grids and serialized pose graphs. These localized map fragments and odometry constraints are continuously transmitted to the central server. Drawing on graph-based SLAM backend optimization principles, the central server analyzes the incoming data to identify geometric overlaps and inter-robot loop closures. Upon detecting a confident match between two distinct local maps, the server establishes virtual constraints between their respective nodes. A nonlinear least-squares optimizer (GTSAM) is then applied to the newly combined pose graph, elastically deforming and aligning the independent trajectories into a single, unified global coordinate frame. This globally consistent 2D occupancy grid is subsequently broadcast to the individual SmartCam units, providing the necessary environmental context for autonomous, decentralized path planning across the campus.

B. Sensor Fusion Pipeline Calibration

To mitigate the inherent challenges of odometric drift, the data pipeline feeds raw telemetry from HiWonder's LiDAR, depth camera, IMU, and wheel encoders to Toolbox's internal GTSAM-based pose graph optimizer. A primary focus of this implementation is developing a Kalman filter to fuse wheel encoder data with IMU measurements. This specifically allows tuning of the covariance matrices, thereby ensuring that the IMU compensates for encoder noise and reduces rotation drift.

This modular approach specifically allows for the sequential integration of sensors, starting with a LiDAR baseline and incrementally adding encoders, IMU, and depth landmarks to better quantify the marginal accuracy gains of each modality. The validation of the sensor fusion layers will be conducted through a comparative analysis of four configurations: LiDAR-only, LiDAR+Encoders, LiDAR+Encoders+IMU, and a full-stack integration that includes depth-camera landmarks. The end goal will therefore be to quantify these results by comparing the robot's estimated trajectory against a physical ground-truth path using the Root Mean Square Error (RMSE).

C. Reinforcement Learning

We developed a Proximal Policy Optimization (PPO)-based local navigation system for the MentorPi robot in a ROS 2-connected Gazebo simulation environment. The policy uses a 30-dimensional observation consisting of 24 LiDAR sectors, the relative goal position and direction, and the robot's measured linear and angular velocities. It directly outputs continuous forward and angular velocity commands. The reward encourages progress toward the goal and successful arrival while penalizing time, unnecessary travel, unsafe proximity to obstacles, and collisions.

Training follows a curriculum that progresses from basic goal reaching and single-obstacle avoidance to randomized clutter and moving-obstacle scenarios. The current mixed curriculum assigns approximately 55% of episodes to complex static scenes, 35% to basic navigation tasks, and 10% to dynamic-obstacle scenes. In the planned integrated system, SLAM will provide a corrected robot pose for computing the relative waypoint given to the PPO policy, allowing the robot to perform local waypoint-to-waypoint navigation while mapping continues online. The next stage is to transfer this policy to the physical MentorPi and reduce sim-to-real differences in LiDAR sensing, robot motion, and control latency.

D. Vision-Language-Action (VLA) Pipeline

A Gazebo and ROS 2 simulation pipeline supports demonstration-based control alongside PPO. A rule-based RGB target-tracking teacher generates bounded differential-drive commands (v, ω), with LiDAR-based slowing and stopping. AutoTeacher records camera, LiDAR, odometry, control commands, and simulation time in ROS bags. Audited, time-aligned observations and actions are converted to LeRobot datasets. SmolVLA [4] was selected for LeRobot/PyTorch compatibility and feasible fine-tuning on an NVIDIA RTX 3070 (8 GB) workstation. The teacher demonstrations use rule-based control; learned VLA predictions are evaluated offline.

E. Communication Architecture: Socket to ROS2

Initial prototyping used a tightly coupled point-to-point socket connection between the Jetson Nano and Raspberry Pi. This was replaced by a ROS2 publish-subscribe architecture in which the Jetson publishes detections to a shared ROS2 Topic, allowing any number of Raspberry Pi subscribers to consume the data, thereby improving scalability and decoupling the perception and control layers. The multi-step pipeline (shown in the multi-layered safety flow) proceeds from obstacle detection, through a UDS socket handoff into a Docker-hosted ROS2 container, through actuation signal generation, and finally to motor stoppage, preserving a clean separation of concerns between the safety layer and the navigation stack.

F. End-to-End Execution Flow

i. Exploration: Car A and Car B will be deployed on different corners of a room. Both start running SLAM Toolbox locally, building their own independent maps.

ii. Local Navigation: Local occupancy maps and SLAM pose estimates provide environmental context and localization for navigation. The PPO controller is currently evaluated as a separate local-navigation branch; in the planned integrated system, the SLAM pose will be used to update the relative waypoint provided to the PPO policy.

iii. Obstacle Avoidance: The robot receives LiDAR, relative-goal, and motion information through ROS 2, and the PPO policy uses these inputs to select forward speed and turning rate. The robot follows these commands and returns updated sensor information until it reaches the goal region, collides, or reaches the step limit.

iv. Data Transmission: Throughout this drive, both cars are streaming their compressed pose-graphs and local map data over the Tailscale VPN to the central Jetson server.

v. Global Optimization: Car A and Car B eventually turn into the same hallway. The central server observes that the LiDAR geometry from Car A matches that of Car B. It creates an inter-robot factor constraint in GTSAM.

vi. VLA Development Branch: AutoTeacher demonstrations are audited and converted to LeRobot for offline SmolVLA training and action prediction. No learned VLA commands have been issued to the robot; held-out evaluation and read-only shadow inference precede any Gazebo closed-loop trials.

vii. The Result: The GTSAM optimizer runs, instantly correcting any odometry drift and merging the two separate grids into one unified campus map. The central server broadcasts this global map back to the fleet, allowing Car A to plot a path into the territory that Car B just finished exploring.

III. Results

The current phase of the SmartCam project has successfully transitioned from the baseline leader-follower architecture into

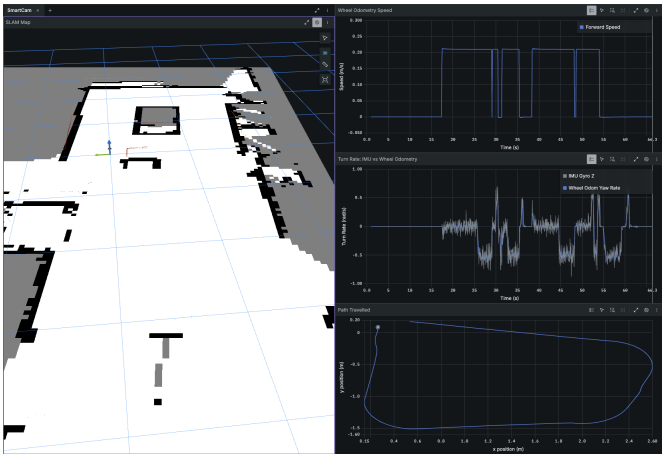


Fig. 2: LiDAR + Wheel Encoder + IMU SLAM Mapping Results and Key Metrics

the formalized design and component validation of a centralized, multi-agent autonomous system. The primary results of this period include finalized architectural blueprints, validated communication topologies, and integrated algorithmic pipelines engineered for deployment across the Rice University campus.

A. Network Communication

A critical bottleneck identified in early iterations was the inability of the standard ROS 2 Data Distribution Service (DDS) to operate on the university's restricted, multicast-blocking wireless network. As a result, a novel communication architecture was designed and validated. The system now uses a Tailscale Virtual Private Network (VPN) overlay in combination with the Zenoh protocol. This configuration successfully bridges ROS 2 nodes using highly compressed, unicast point-to-point connections.

B. SLAM and Sensor Fusion Baseline

Initial testing has thus far focused on establishing a stable data pipeline and characterizing raw sensor noise. To improve the robustness and accuracy of the mapping, we integrate sensor data using a sensor fusion pipeline. This design helps compensate for the limitations of individual sensors, such as LiDAR sparsity and noise in dynamic environments. While full-scale mapping is currently underway, SLAM was incrementally validated across three sensor configurations. Figure 2 illustrates the baseline mapping and sensor performance across the integrated pipeline. The SLAM-generated map is displayed on the left, while the right panel shows the corresponding kinematic and state-estimation metrics across three subplots.

From the top-right plot, the wheel encoder pipeline was evaluated at a sampling rate of $4\,\text{Hz}$ across four distinct motion bursts totaling $32.1\,\text{s}$ of active trajectory runtime ($13.0\,\text{s}$, $4.0\,\text{s}$, $9.7\,\text{s}$, and $5.4\,\text{s}$). Driven at a constant forward wheel velocity of $v = 0.21\,\text{m/s}$, the estimated distance travelled was calculated as follows:

$$d_{\text{calc}} = v \times t = 0.21\,\text{m/s} \times 32.1\,\text{s} = 6.741\,\text{m} \quad (265.39\,\text{in}) \quad (1)$$

In the middle-right plot, the rotational velocity profiles show strong agreement between the IMU gyroscope and the wheel-odometry calculation. This alignment validates the accuracy of the underlying kinematics, specifically the wheel-radius and track-width calibrations. The transient spikes and high-frequency noise observed in the raw IMU signal primarily stem from motor-induced vibrations transmitted through the chassis. As a next step, blending these complementary sources in an Extended Kalman Filter (EKF) should attenuate high-frequency chassis noise and mitigate drift caused by wheel slippage during acceleration.

Finally, the bottom-right plot depicts the robot's reconstruction path. Comparing this estimated trajectory against the physical ground truth perimeter loop of $269.0\,\text{in}$ ($6.833\,\text{m}$, representing an $83\,\text{in} + 50\,\text{in} + 83\,\text{in} + 50\,\text{in}$ rectangular test cell) yields an absolute trajectory length error of $3.61\,\text{in}$ ($0.092\,\text{m}$), corresponding to an Absolute Percentage Error (APE) of 1.346%:

$$\text{APE} = \left| \frac{269.0 - 265.39}{269.0} \right| \times 100\% = 1.346\% \quad (2)$$

The next step is to further improve the sensor data for maximum accuracy. This will involve tuning the Kalman Filter to establish a reliable baseline for position and orientation. Following this, the depth camera will be integrated to provide visual landmark constraints. The final validation of this phase will produce a comparative performance table quantifying key metrics for the overall system pipeline.

C. Reinforcement Learning

The PPO navigation system was evaluated on a fixed baseline set containing seven scenario categories with two episodes per category, for a total of 14 episodes. The current baseline reached the goal in all 14 trials, including single-obstacle, multi-obstacle, direct-path, offset-obstacle, and slow dynamic-obstacle cases. More difficult cluttered and dynamic evaluations remain less reliable, with selected recent checkpoints reaching 5/6 successful episodes on some six-episode test sets. These results indicate that the basic navigation behavior is established, while robustness in dense and dynamic environments remains an area for further improvement.

We also explored how the number of LiDAR sectors affects navigation in one- and two-obstacle simulations. We selected 24 sectors as the current policy input because training converged with this resolution. Tables I and II summarize the results for the different sector counts; the 48-sector evaluation is still in progress.

TABLE I: LiDAR-sector results in the one-obstacle simulation. C/T denotes collisions/timeouts. The final column lists mean path length, final goal distance, and clearance (m), respectively.

Sectors	Success	C/T	Steps	Path / final dist. / clearance (m)
24	5/5	0/5, 0/5	76.4	3.173 / 0.236 / 0.371
12	5/5	0/5, 0/5	75.6	3.250 / 0.229 / 0.414
8	2/5	3/5, 0/5	54.6	2.011 / 0.953 / 0.168
6	2/5	3/5, 0/5	56.2	2.102 / 0.945 / 0.237
4	0/5	1/5, 4/5	101.2	4.059 / 0.793 / 0.548

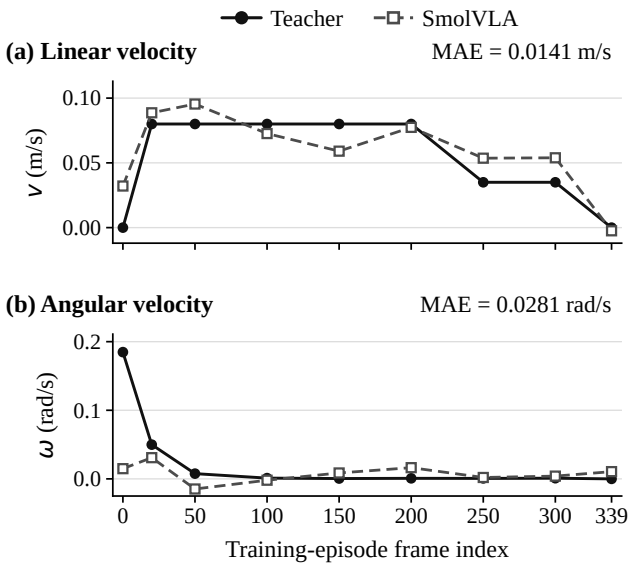


Fig. 3: Rule-based teacher commands and SmolVLA offline predictions on nine frames from the same training episode: (a) linear velocity and (b) angular velocity. Dashed curves connect sampled predictions; this is an in-sample action comparison, not a closed-loop navigation result.

TABLE II: LiDAR-sector results in the two-obstacle simulation. C/T denotes collisions/timeouts. The final column lists mean path length, final goal distance, and clearance (m), respectively.

Sectors	Success	C/T	Steps	Path / final dist. / clearance (m)
24	4/5	0/5, 1/5	110.2	4.646 / 0.250 / 0.262
12	4/5	0/5, 1/5	110.4	4.691 / 0.255 / 0.301
8	3/5	1/5, 1/5	97.4	4.044 / 0.991 / 0.323
6	3/5	1/5, 1/5	98.2	4.064 / 1.040 / 0.371
4	0/5	1/5, 4/5	101.4	4.043 / 1.781 / 0.406

D. VLA Offline Evaluation

One *partial* blue-target demonstration yielded 340 frames at 10 Hz. SmolVLA was fine-tuned for 100 BF16 steps with batch size 1 and a frozen vision encoder; the resulting checkpoint was successfully reloaded. On nine sampled frames from the *same training episode* (indices 0, 20, 50, 100, 150, 200, 250, 300, and 339), linear and angular velocity MAEs relative to teacher commands were 0.0141 m/s and 0.0281 rad/s, respectively (Fig. 3). Initial turning and stopping remain error modes.

These are *in-sample offline action imitation results*, not evidence of successful closed-loop VLA navigation, generalization, or physical robot deployment. Goal completion was not verified, and the trained VLA has not controlled the simulated or physical robot.

E. Centralized map merging architecture

To achieve multi-agent environmental awareness without exceeding the computational limits of the embedded hardware, a 2D pose-graph map-merging strategy is formalized. Rather than transmitting bandwidth-heavy 3D point clouds, the edge rovers use local SLAM algorithms to generate highly compressed 2D occupancy grids and serialized odometry graphs. The computational load of map merging will be offloaded to a central Jetson Orin Nano hub. This central server is tasked with applying graph-based backend optimization to detect geometric overlaps, establish inter-robot constraints, and elastically deform the independent local graphs into a single, unified global coordinate frame for campus-wide path planning.

IV. Discussion

SmartCam develops a low-cost architecture for SLAM, sensor fusion, communication, and learning-based control. Component results should be distinguished from the proposed integrated autonomy pipeline. The transition from a tightly coupled socket architecture to a scalable ROS2 publish-subscribe model resolved the primary network communication bottleneck imposed by the university's restricted wireless infrastructure and positioned the system to scale to additional rover units with minimal reconfiguration.

The low APE of 1.346% directly validates the physical system design, confirming that the wheel encoder geometry, motor encoders, and raw kinematic models are inherently accurate even prior to heavy software filtering. The incremental sensor-fusion results confirm that integrating wheel-encoder odometry and IMU signals with LiDAR-only SLAM provides a highly accurate foundation. Conducting a comprehensive quantitative comparison across all four sensor configurations remains the key next step to rigorously validate these qualitative improvements and to determine the marginal accuracy gain from each additional modality.

The VLA limitations in Section II-D motivate explicit teacher termination records, multiple audited demonstrations, episode-level held-out tests, and read-only shadow inference before safeguarded Gazebo closed-loop evaluation.

For reinforcement learning, the current simulation results demonstrate reliable navigation in basic and medium-difficulty environments, but performance in dense clutter and dynamic scenes remains less consistent. The next steps are to improve robustness in these scenarios, integrate SLAM-based pose estimates for waypoint navigation, and transfer the PPO controller from Gazebo to the physical MentorPi while characterizing and reducing the sim-to-real gap.

Three priorities constitute the primary future work: (i) integrating depth camera data into the GTSAM pose graph to begin map merging; (ii) transferring the RL policy from simulation to the real robot and integrating SLAM-based localization with waypoint-to-waypoint RL navigation; and (iii) AutoTeacher validation, held-out VLA evaluation, and read-only Gazebo shadow inference. Physical VLA deployment and integrated campus navigation remain future work.

Credit Author Statement

All authors contributed equally as collaborators and shared responsibility throughout the project. Each author played a primary role while also supporting other aspects of the work as part of a horizontal and cooperative team effort.

- **Ashley Garcia:** Conceptualization, SLAM Pipeline & GTSAM Optimization, Map Merging, Methodology, Sensor Fusion, Visualization, Validation, Writing : Original Draft, Cross-Team Coordination
- **Chen-En Lin:** Software Development, Hardware Integration, Reinforcement Learning, Testing, Writing : Review & Editing
- **Chenghao Xiang:** Conceptualization, Gazebo AutoTeacher, ROS-to-LeRobot Data Pipeline, SmolVLA BF16 Fine-Tuning, Offline Evaluation, Hardware Integration, Methodology, Visualization, Cross-Team Coordination
- **Youjia Tong:** Software Development, Wireless Communication Validation, Experimental Investigation, Validation, Obstacle Detection Integration, Reinforcement Learning, Testing, Writing : Review & Editing

All authors participated in discussions, reviewed the manuscript, and approved the final version for publication.

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